

Note 12.3 The Dot Product

Angle Between Vectors

Definition of Dot Product

The **dot product** $\mathbf{u} \cdot \mathbf{v}$ of vectors $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ is

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3 = |\mathbf{u}| |\mathbf{v}| \cos \theta$$

where θ is the angle between the two vectors.

Theorem 1--Angle Between Two Vectors

The angle θ between two nonzero vectors $\mathbf{u} = \langle u_1, u_2, u_3 \rangle$ and $\mathbf{v} = \langle v_1, v_2, v_3 \rangle$ is given by

$$\theta = \cos^{-1} \left(\frac{u_1v_1 + u_2v_2 + u_3v_3}{|\mathbf{u}| |\mathbf{v}|} \right)$$

or in the notation of the dot product,

$$\theta = \cos^{-1} \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}| |\mathbf{v}|} \right)$$

The proof can be done by using the law of cosines, constituting the vector $\mathbf{w} = \mathbf{u} - \mathbf{v}$.

Perpendicular (Orthogonal) Vectors

Definition

Vectors $\mathbf{u}$ and $\mathbf{v}$ are **orthogonal** (or **perpendicular**) if and only if $\mathbf{u} \cdot \mathbf{v} = 0$. And the **zero vector** is perpendicular to any vector.

Dot Product Properties and Vector Projections

Properties of the Dot Product

If $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are any vectors and c is a scalar, then

- $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
- $(c\mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (c\mathbf{v}) = c(\mathbf{u} \cdot \mathbf{v})$
- $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$
- $\mathbf{u} \cdot \mathbf{u} = |\mathbf{u}|^2$

- $\mathbf{0} \cdot \mathbf{u} = 0$

The upper properties can be proved by the definition of the dot product.

Vector Projection

The **scalar component** of $\mathbf{u}$ in the direction of $\mathbf{v}$ is the scalar

$$|\mathbf{u}| \cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|} = \mathbf{u} \cdot \frac{\mathbf{v}}{|\mathbf{v}|}$$

The **vector projection** of $\mathbf{u}$ onto $\mathbf{v}$ is the vector

$$\text{proj}_{\mathbf{v}} \mathbf{u} = |\mathbf{u}| \cos \theta \frac{\mathbf{v}}{|\mathbf{v}|} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \right) \mathbf{v}$$

Exercises 12.3

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